

NYKAA PRODUCT ANALYSIS USING WEB SCRAPING, DATA CLEANING & EXPLORATORY DATA ANALYSIS (EDA)

→Abstract

The rapid growth of e-commerce has generated a vast amount of product data that can be analyzed to understand customer preferences, pricing strategies, discounts, and product performance. This project focuses on collecting product information from the Nykaa website using web scraping techniques, cleaning and analyzing the data, and creating an interactive Power BI dashboard.

The dataset includes details such as product names, brands, categories, prices, discounts, ratings, and customer reviews. After preprocessing the data, exploratory data analysis (EDA) was performed to identify trends and patterns. Finally, an interactive Power BI dashboard was developed to visualize key insights, enabling better decision-making and product analysis.

→Introduction

Nykaa is one of India's leading beauty and personal care e-commerce platforms. It offers thousands of products from various brands across multiple categories. Since product information changes frequently, web scraping provides an efficient way to collect up-to-date product data for analysis.

This project demonstrates the complete data analytics workflow:

1. Web scraping
2. Data cleaning
3. Exploratory Data Analysis (EDA)
4. Dashboard creation using Power BI

The project helps understand pricing patterns, product distribution, discounts, customer ratings, and brand performance.

→Business Understanding

Nykaa is one of India's leading beauty and personal care e-commerce platforms. Understanding customer preferences, pricing strategies, discounts, and product ratings helps businesses improve product positioning, optimize pricing, and enhance customer satisfaction. This project analyzes Nykaa product data to identify trends that support better business decisions.

→Problem Statement

Online shopping platforms contain a large amount of product information, making it difficult to manually analyze market trends. Businesses and analysts need an automated approach to collect, process, and visualize this data.

The objective of this project is to scrape product data from Nykaa, analyze it, and build an interactive dashboard that provides meaningful business insights.

→Objectives

- Collect product information from the Nykaa website using web scraping.
- Store the scraped data in a structured dataset.
- Clean and preprocess the collected data.
- Perform exploratory data analysis.
- Analyze pricing, ratings, brands, and product categories.
- Create an interactive Power BI dashboard.
- Generate business insights to support decision-making.

→Tools and Technologies

Tool	<u>Purpose</u>
Python	Web scraping and data analysis
Beautiful Soup	HTML parsing
Requests	Accessing web pages
Pandas	Data cleaning and analysis
NumPy	Numerical operations
Matplotlib	Data visualization
Power BI	Dashboard development
Excel/CSV	Data storage
Statistical	Data Visualization

→Methodology

Phase 1: Web Scraping

- Access the Nykaa product pages.
- Extract product details.
- Store the extracted information in CSV format.

Data Collected

- Product Name
- Brand
- Category
- Price
- Discount
- Rating
- Number of Reviews

→Example:

- Website: Nykaa
- Library Used: Requests and BeautifulSoup
- Total Rows Scraped: (mention your actual number)
- Total Columns: (mention your actual number)
- File Format: CSV

Phase 2: Data Cleaning

The collected data was cleaned using Python and Pandas.

Cleaning included:

- Removing duplicate records
- Handling missing values
- Correcting data types
- Standardizing column names
- Removing unwanted symbols
- Converting prices into numeric values

Phase 3: Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset.

Analysis included:

- Product distribution by category
- Brand-wise product count
- Price analysis
- Rating analysis
- Discount analysis
- Review analysis

Phase 4: Dashboard Development

Power BI was used to create an interactive dashboard.

The dashboard provides:

- KPI Cards
- Charts
- Filters
- Interactive analysis

→Dataset Description

The dataset consists of product information collected from the Nykaa website.

Example fields include:

Column	Description
• Product Name	Name of the product
• Brand	Product brand
• Category	Product category
• Price	Selling price
• Discount	Discount percentage

- Rating
- Reviews

Customer rating

Number of customer reviews

→Exploratory Data Analysis

The analysis focused on identifying:

- Most popular product categories
- Highest-rated brands
- Products with maximum discounts
- Average product prices
- Customer review trends
- Price distribution

→Power BI Dashboard

The dashboard includes:

KPI Cards

- Total Products
- Total Categories
- Average Price
- Average Rating
- Total Reviews
- Average Discount

→Visualizations

- Products by Category
- Top Brands
- Average Price by Category
- Category Distribution
- Rating Distribution
- Discount Analysis

- Reviews by Brand
- Price vs Rating Scatter Plot

→Interactive Features

- Category filter
- Brand filter
- Price filter
- Rating filter

→Key Insights

- Some brands offer significantly more products than others.
- Product prices vary widely across categories.
- Customer ratings are generally concentrated in the higher range.
- Discounts differ by product category and brand.
- A few brands receive the majority of customer reviews.
 - ➔ Instead of general statements, relate them to your dataset. For example:
- Beauty products account for the largest share of the dataset.
- Premium brands generally have higher average prices.
- Products with higher ratings often receive more reviews.
- Discount percentages vary across categories.

→Advantages

- Automated data collection
- Reduced manual effort
- Easy visualization of product data
- Interactive reporting
- Better understanding of market trends
- Supports business decision-making

→Limitations

- Website structure changes may affect scraping scripts.
- Data reflects only the scraping period.
- Some product information may be unavailable or missing.
- Dynamic content may require additional scraping techniques.

→Future Scope

- Integrate live data updates.
- Add sales prediction using machine learning.
- Include sentiment analysis of customer reviews.
- Compare products across multiple e-commerce platforms.
- Publish dashboards using the Power BI Service.

→References

- Nyka Official Website
- Python Documentation
- Pandas Documentation
- Beautiful Soup Documentation
- Power BI Documentation

→Conclusion

This project demonstrates a complete data analytics workflow, starting with web scraping product data from Nykaa and ending with an interactive Power BI dashboard. Through data cleaning, exploratory analysis, and visualization, the project provides meaningful insights into product categories, brands, pricing, discounts, ratings, and customer reviews. The dashboard enables users to explore the data interactively and supports informed decision-making for business and market analysis.

